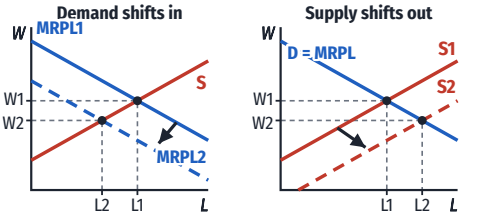
Symbol key. W wage · L workers · MPL marginal product of labor · MR marginal revenue (= P for a competitive firm) · MRPL = MR × MPL · MPK, MRPK the same for capital K · R rental price of capital · P price · Q quantity · D demand · S supply (or national saving) · Pw world price · QS, QD quantity supplied, demanded · t tariff per unit · OC opportunity cost · ToT terms of trade · PPC, CPC production, consumption possibilities · WM washing machines · Y real GDP · Y* potential output · C consumption · I investment · G government purchases · NX net exports = EX (exports) − IM (imports) · T net taxes (taxes minus transfers) · PAE planned aggregate expenditure · mpc marginal propensity to consume · A autonomous spending · i nominal interest rate · r real interest rate · π inflation · RF Fed reaction function · e price of $1 in another currency · NKI net capital inflows · MB, MC, MU marginal benefit, cost, utility · EMC, SMC, PMC external, social, private marginal cost · DWL deadweight loss · CS consumer surplus · Δ "change in" · 1 before, 2 after

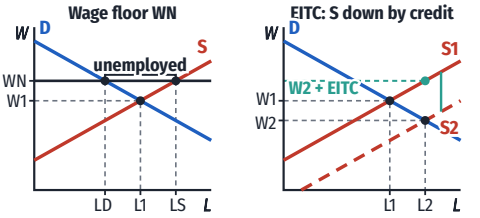
1 Labor market and its shifts — Class 12



1. W up, L across. D = MRPL slopes down; S slopes up. Mark W1 and L1.
2. Shift only the curve the story touches; mark W2, L2, and an arrow.

- MR or MPL falls (trade war, cheaper imports, machines replace these workers): D shifts in, so W and L fall. A rise shifts D out.
- Immigration shifts S out: W falls, L rises. Removals shift S in, and partners of those removed lose MPL, so their D shifts in.
- Reverse: W and L moved the same way means D shifted; opposite ways means S shifted.

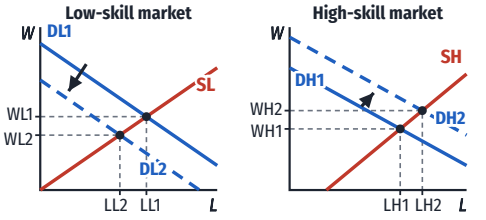
2 Wage floor and the EITC — Classes 12, 13



1. Floor: a flat line at WN above W1. Firms hire LD (on D); LS want work (on S).
2. EITC: S shifts down by the credit. Employers pay W2; workers get W2 + EITC.

- Unemployment = LS − LD. D: W = 30 − 2L, S: W = 6 + 2L, WN = 22 give LD = 4, LS = 8, so 4 are unemployed.
- A floor below W1 does nothing. The EITC raises L, and W2 < W1 < W2 + EITC. A subsidy paid to employers shifts D up instead.

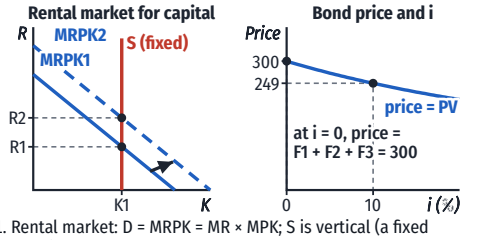
3 Two labor markets: inequality — Class 13



1. Low-skill panel on the left, high-skill on the right, WH1 above WL1. Shift only the curves the story names; inequality is the gap WH − WL.

- Globalization (shown): low-skill goods get cheaper, so DL shifts in; high-skill export prices rise, so DH shifts out. The gap grows.
- Skill-biased technology: only DH shifts out. More college graduates: SH shifts out and SL shifts in, so the gap shrinks.

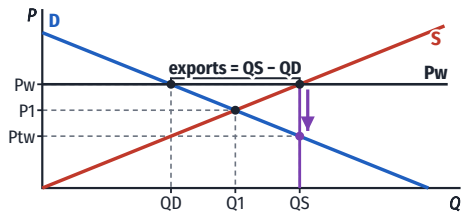
4 Capital: rental price and bond price — Class 14



1. Rental market: D = MRPK = MR × MPK; S is vertical (a fixed stock). A higher output price raises only R. More machines lower R and raise K.
2. Bond: the price falls as i rises. At i = 0 it is F1 + F2 + F3.

- $100 a year for 3 years is worth $300 at i = 0 and $248.69 at i = 10%.
- Lower future MPK or MR lowers each MRPK, so the price of capital falls.

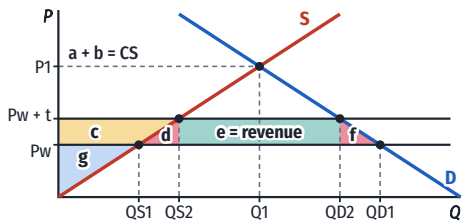
5 Exports, then a trade war — Class 16



1. Mark P1 and Q1, then a flat Pw above P1. Read QD on D and QS on S; exports = QS − QD.
2. Trade war with a fixed harvest that cannot be stored: go straight down from QS to D. That height is Ptw, below P1. Next season P drifts back toward P1.

- D: P = 12 − Q, S: P = Q, Pw = 8 give QD = 4, QS = 8, exports 4. Trade war: Ptw = 12 − 8 = 4 (P1 was 6).
- If other buyers pay more than Ptw, their price is the floor. Reverse: exports of 6 need Pw − (12 − Pw) = 6, so Pw = 9.

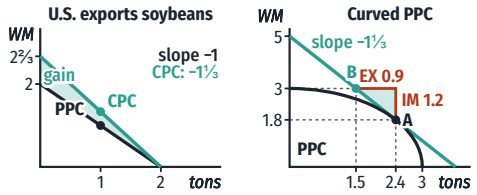
6 Imports with a tariff or a quota — Class 16



1. Flat Pw below P1; imports = QD1 − QS1.
2. Draw Pw + t; read QS2 on S and QD2 on D.
3. Letter the strip between the lines c, d, e, f (g is producer surplus before).

- Buyers lose c + d + e + f. Producers gain c. Revenue e = t × (QD2 − QS2). DWL = d + f.
- D: P = 20 − Q, S: P = Q, Pw = 4, t = 2 give c = 10, d = 2, e = 16, f = 2.
- Quota: draw "S + quota" (S shifted right by the quota above Pw). Equal imports give the tariff's price; e becomes quota rent.

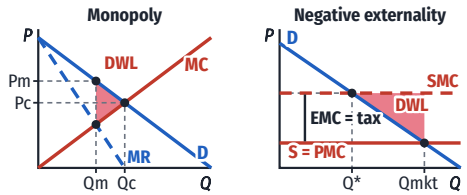
7 PPC and CPC: gains from trade — Class 15



1. Straight PPC: the intercepts are one worker's output (U.S. 2 tons or 2 WM; China 1 ton or 2 WM). The CPC starts at the export good with slope −ToT.
2. Bowed-out PPC (rising OC): make at A, where a line with slope −ToT touches it; use at B. The country makes both goods.

- ToT = export price ÷ import price = $400 ÷ $300 = 1⅓ WM per ton, between the OCs 1 and 2. China's ToT is the reciprocal, ¾.
- A better ToT pivots the CPC out. Reverse: ToT = CPC vertical intercept ÷ horizontal intercept.
- Gain at 1 ton: CPC 1⅓ − PPC 1 = ⅓ WM more.

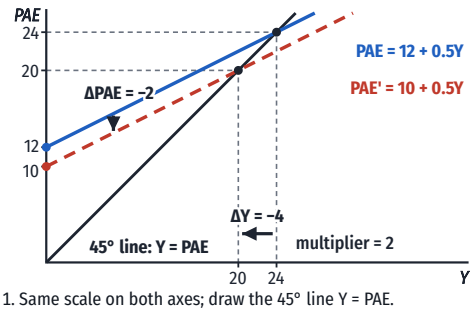
8 Monopoly and negative externality — micro



1. Monopoly: D = MB. MR starts with D and is twice as steep. Qm is where MR = MC; go up to D for Pm.
2. Externality: SMC = PMC + EMC. The market makes Qmkt (D meets PMC); the efficient Q* is where D meets SMC. A tax equal to EMC moves it to Q*.

- A monopolist raises P and cuts Q. Example: P = 300 − 10Q, MC = 10Q: 10 units at $200 against 15 at $150.

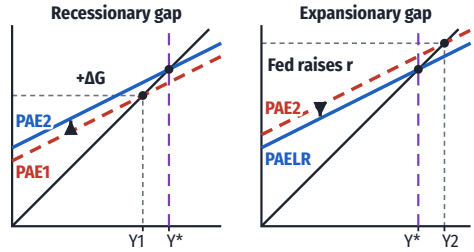
9 Keynesian cross: base and shock — Classes 19 to 21



1. Same scale on both axes; draw the 45° line Y = PAE.
2. PAE = C + I + G + NX starts at A and rises with slope mpc.
3. The crossing is Y = A ÷ (1 − mpc). Mark it on both axes.
4. Shock: PAE' is parallel, moved by ΔPAE. Mark its intercept and its crossing.

- 12 + 0.5Y gives Y = 12 ÷ 0.5 = 24; 10 + 0.5Y gives 20.
- ΔPAE = −2 is the vertical shift. ΔY = 2 × (−2) = −4 is the horizontal move.
- Right of the crossing Y > PAE, so unsold goods pile up and firms cut output.

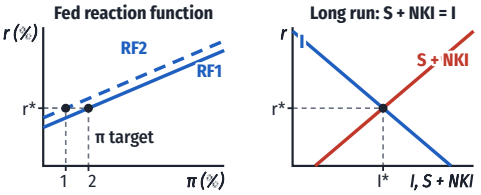
10 Output gaps and the policy fix — Classes 20 to 22



1. Draw a dashed line at Y*. Recessionary gap (Y1 < Y*): raise G, cut T, or the Fed cuts r. PAE shifts up until it crosses at Y*.
2. Expansionary gap (Y2 > Y*): inflation creeps up, so the Fed raises r. PAE shifts down to PAE_{LR} (long run) at Y*.

- Needed ΔPAE = (Y* − Y) × (1 − mpc). Example: Y = 20, Y* = 24, mpc = 0.5: raise G by 2, or cut T by 2 ÷ 0.5 = 4.

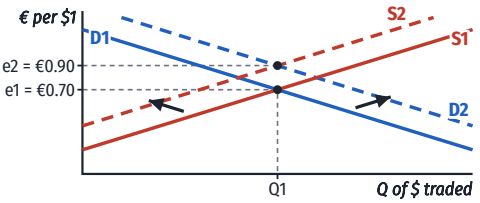
11 Fed reaction function, long-run r* — Classes 22, 23



1. RF slopes up: higher π, higher r. At r*, read the π target.
2. Lower long-run inflation: shift RF up to RF2. After a short recession, π settles at the lower target (2% to 1% here).
3. Long run: I slopes down, S + NKI slopes up. Monetary policy moves neither, so r* stays put.

- RF: r = 3 + 0.5 × (π − 2). At π = 4%, r = 4% and i = r + π = 8%.
- Bigger budget deficits shift S + NKI left: r* rises, I falls, NX falls. A permanent rise in spending also raises r*.

12 Market for dollars — Class 23



1. D is foreigners buying U.S. goods and assets. S is Americans buying foreign ones.
2. Shift, then mark e2. If e2 is above e1, the dollar appreciated.

- U.S. real rate rises (shown): D right, S left, so the dollar appreciates and NX falls.
- Foreign rate rises: D left, S right, the dollar depreciates. Safe-haven rush: D right. U.S. tariffs: S left.

How to answer (his grading)

- Answer first with the direction word and the number ("decreased by $2 trillion"), then the reason in one to three sentences. Less is more.
- On a graph, label both axes, every curve and intercept, and both coordinates of each equilibrium. On a shift, name the curve, the direction, and the reason through a formula.

Keynesian cross — graphs 9, 10

PAE = A + mpc × Y, where A = C̄ − mpc × T + I + G + NX (the intercept) and C̄ is autonomous consumption. Y = A ÷ (1 − mpc).

Multiplier = 1 ÷ (1 − mpc); ΔY = multiplier × ΔPAE; gap = Y − Y*.

- Short run: prices are sticky, so output follows planned spending. Long run: output returns to Y*, set by labor, capital, and technology.
- Multipliers: mpc 1/3 gives 1.5; 0.5 gives 2; 0.6 gives 2.5; 2/3 gives 3; 0.75 gives 4; 0.8 gives 5; 0.9 gives 10.

If asked	Do this
ΔPAE, then ΔGDP.	ΔPAE adds the changes in the pieces not multiplied by Y, with signs (NX from −1 to 0 is +1); it is the vertical shift. ΔY = multiplier × ΔPAE, the slide along the 45° line. ΔY is larger whenever mpc > 0, because spending becomes income and part of it is spent again.
Offset a shock, or close a gap.	Offset: ΔG = −ΔPAE of the shock, which returns Y to where it was. Close a gap: ΔG = (Y* − Y) × (1 − mpc), which moves Y to Y*. They match only if Y started at Y*.
Use taxes or transfers.	ΔT = −needed ΔPAE ÷ mpc, so needed 2 with mpc 0.5 means cut T by 4. A transfer is a cut in T. Tax changes must be bigger because part is saved.
Equal ΔG and ΔT.	ΔPAE = (1 − mpc) × ΔG, so ΔY = ΔG exactly.
Fed move that closes a gap.	Points of r = needed ΔPAE ÷ spending change per point. Gap 2 with mpc 0.6 needs 0.8; at 0.4 per point, cut r by 2 points. Then i = r + π.
r that puts Y at Y*.	Set the intercept (a − b × r) equal to Y* × (1 − mpc) and solve for r.
Find the mpc.	mpc = ΔC ÷ Δ(Y − T), never C ÷ Y. From two outcomes, mpc = 1 − ΔPAE ÷ ΔY. From a graph, 1 − mpc = intercept ÷ Y, so 12 and 30 give 0.6.
Missing I or G.	The intercept that gives Y* is Y* × (1 − mpc). Subtract the pieces you know.
A table of Y and PAE.	Equilibrium is the row where PAE = Y. mpc = ΔPAE ÷ ΔY between two rows.
Find ΔC.	ΔC = ΔC̄ + mpc × ΔY. If only I moved, ΔC = mpc × ΔY.
New mpc.	The line tilts around the same intercept; recompute A ÷ (1 − new mpc).
Imported spending.	Only the part spent on domestic goods raises PAE.

- **TRAP** Offset a shock with −ΔPAE, never −ΔGDP. With taxes inside C, subtract mpc × T, not T: 6 + 0.5(Y − 2) = 5 + 0.5Y.
- **TRAP** A change in Y moves along PAE. C̄, I, G, NX, T, and r shift it in parallel; the 45° line never moves.

Fiscal policy words — Class 20

- Surplus = T − G. A deficit is one year's shortfall; the debt is all past deficits minus surpluses. Expansionary policy lowers T − G; countercyclical policy fights the current gap.
- Tax multiplier = −mpc ÷ (1 − mpc), one smaller in size than the G multiplier. A tax cut for high-mpc households moves PAE more.
- Debt brings higher interest rates, the risk of a debt crisis, and a burden on future generations. Run surpluses in good times.
- A balanced-budget rule cuts G in a recession and deepens it. Stimulus can be too small, too big, or too late; the Recovery Act offset part of the fall (Y2 < Y3 < Y*).
- G rose but Y did not change: other spending fell by as much.
- Automatic stabilizers: taxes fall and transfers rise in a recession with no new law. The structural deficit is the deficit at Y*.

Long-run growth — Class 18

- $Y_t = Y_0 \times (1 + g)^t$. Average yearly growth = $(Y_t \div Y_0)^{1/t} - 1$, so 100 to 121 in 2 years is 10%.

Rule of 70: years to double ≈ 70 ÷ g%, and g ≈ 70 ÷ years. Eightfold is 3 doublings. $1.01^{70} \approx 2$ and $1.01^{200} \approx 7.3$.

- Growth per person ≈ GDP growth − population growth. Catching up from half takes about 70 ÷ ($g_B - g_A$) years.
- Y/POP = (Y/N) × (N/POP), with N workers and POP population. $30,000 per person and $75,000 per worker mean N/POP = 0.4.
- Over time, technology explains most growth (about 2/3). Across countries, capital and technology explain about half each; the technology gap is mostly institutions.
- Roads are physical capital, schooling is human capital, and methods and institutions are technology. Social infrastructure: rule of law, property rights, protection from government.
- Inventions help others, so markets make too few. Fixes: education, R&D subsidies, secure property rights.
- Growth rising from 1% to 2% is a rise of 1 percentage point, or 100%.

The Fed, inflation, banks — graphs 10, 11

The Fed sets i (monetary policy); Congress sets G and T (fiscal policy). r = i − π, and π moves slowly, so r moves with i.

- Unexpected inflation lowers the real rate a loan pays, so it helps borrowers and hurts lenders.

If asked	Do this
The Fed raises r.	I, C (new cars), and NX (the dollar rises) fall, so PAE shifts down and Y falls by the multiplier.
π up, i fixed.	r falls, so policy got looser with no Fed move.
Open-market purchase.	The Fed creates money and buys bonds: reserves rise, the fed funds rate falls, bond prices rise. It is not G. Higher interest on reserves raises the fed funds rate.
Y is not at Y*.	Right away π does not move (menu costs, contracts, limited information); at Y* it stays at expected inflation. Over time π rises if Y > Y* and falls if Y < Y*, and the Fed moves r until Y = Y*. Reverse: π falling for years means Y < Y*.
His rule r = 3 + 0.5(π − 2).	π = 4 gives r = 4 and i = 8. In nominal terms i = 2 + 1.5π, so π up 2 points raises i 3 points, and i = 9.5 means π = 5.
Deflation.	π = −2 needs r = 1, so i = −1. Rates cannot go far below 0 (cash pays 0), so the Fed uses forward guidance and quantitative easing.
Long run.	Y = Y*, r = r*, and π is where the RF crosses r*, not necessarily 2%. RF r = 1 + 0.5π and r* = 3 give π = 4%.
Along the RF, or a shift?	π moved and the Fed followed its rule: a movement. A new rule is a shift, the only way to change long-run π. A steeper rule deepens the short-run recession.
Oil price jump.	π rises at once (an inflation shock). The Fed raises r along its RF, so Y dips below Y*, then returns.
Why did the Fed move?	To offset another force (2008, COVID), to cut inflation, or by mistake (the Depression, when tight money raised r).
Bank solvency.	Equity = loans + reserves + bonds − deposits − borrowing; negative is insolvent. Rate rises hit long-term bonds hardest.
Runs, contagion.	A run forces fire sales that can sink a sound bank. Channels: confidence, linkage (a loan between banks), fire sale, macroeconomic.
Financial crisis.	Wider credit spreads (risky rate − safe rate), tighter lending, and lower confidence shift PAE down. Prevent with capital requirements and deposit insurance.

- **TRAP** "Inflation falls" means prices rise more slowly, not that prices fall.

Exchange rates, capital flows — graph 12

Ask whose price the number is: ¥ per $1 prices a dollar, $ per €1 prices a euro. A higher price means that currency appreciated.

- Percent changes use each currency's own old price: ¥160 to ¥150 per $1 is −6.25% for the dollar and +6.67% for the yen.
- Foreign price = dollar price × rate, so $50 at ¥150 is ¥7,500. Dollar price = foreign price ÷ rate. Flip a quote with 1 ÷ rate.
- A foreign recession shifts D for dollars left, so the dollar depreciates. When D and S shift in opposite directions, the price change is certain but the quantity is not. A strong dollar helps importers and Americans abroad and hurts exporters.

Long run: NX = −NKI and S + NKI = I, so NKI = I − S; r* is the real rate where they balance. A trade deficit means saving is below investment.

- Tariffs move the dollar but not NKI, so NX stays the same. A higher U.S. rate raises NKI and lowers NX. Bad U.S. goods or a taste for imports would weaken the dollar, not cause a deficit.
- Balance of payments: EX + KI = IM + KO (KI, KO are capital inflows, outflows). EX 3, IM 4, KO 1 gives KI = 2.
- Law of one price: e = foreign price ÷ U.S. price. It holds for traded goods like gold, not for Big Macs (local labor and rent). A fixed rate above the market rate forces the central bank to buy its currency with reserves.

Capital and present value — graph 4

PV = F ÷ $(1 + i)^t$ with i as a decimal; add the PV of each payment. Price of capital = the PV of future MRPKs.

- When rates rise, bond, stock, and machine prices fall, long-term bonds most. Bond prices fell means rates rose.
- Implied rate: 1 + i = F ÷ price, so $1,050 next year for $1,000 is 5%. Paid forever, PV = F ÷ i: $50 a year at 5% is $1,000.
- Buy a machine if the PV of its MRPKs is at least its price: $1,000 a year for 3 years at 5% is $2,723.25.
- Investment demand: a change in i is a move along it. A tax credit shifts it right; a higher machine price, lower MPK, or pessimism shifts it left.
- Lower rates raise stock prices two ways: a higher PV of dividends and higher expected dividends.
- "Capital" covers machines, financial assets, and savings that cross borders. Investment counts only new physical capital.

GDP, prices, jobs — Class 17

Y = C + I + G + NX. I is new capital, new housing, and inventory change. G excludes transfers.

- Count final goods newly made here this period, or add each stage's value added. Used goods, stocks, and transfers are out. An import raises C and IM equally, so Y is unchanged. Paying a housekeeper adds to GDP.
- CPI = cost of the base-year basket now ÷ its base-year cost × 100. π = change in the index ÷ last year's index.

$X of year A in year-B dollars = X × $CPI_B \div CPI_A$. Real price change = (new ÷ old price) ÷ (new ÷ old CPI); above 1, it rose.

- Real GDP = nominal ÷ deflator × 100; the two are equal in the base year. Real growth ≈ nominal growth − inflation.
- The CPI includes imports and the deflator does not. The CPI overstates living costs (quality, substitution, new goods).
- Core CPI drops food and energy. GDP per person beats the typical wage because it counts capital income. PPP compares countries at one set of prices.
- Unemployment rate = unemployed ÷ labor force (employed + unemployed). Participation = labor force ÷ adults. The unemployed looked for work in the last 4 weeks; they are frictional, structural, or cyclical.
- Discouraged workers leave the labor force: 8 ÷ 160 = 5% becomes 6 ÷ 158 = 3.8% with no new jobs.

Labor and inequality — graphs 1 to 3

If asked	Do this
Hire from a table.	MPL = change in output and MRPL = P × MPL. Hire every worker with MRPL ≥ W. With market power, MRPL is the change in total revenue.
MR and MPL both move.	Multiply the factors: MPL × 1.5 with price × 0.6 gives 0.9, so demand shifts in.
Solve with numbers.	D = S: 30 − 2L = 6 + 2L gives L = 6, W = 18. "3 fewer at every wage": put L + 3 in S, so L = 4.5, W = 21.
Mariel boatlift.	Cubans came for political reasons, a natural experiment. Wages barely fell: labor demand may be very elastic.
Fractiles, shares.	P90-95 is the 90th to 95th percentile ($153,000 to $233,000 in 2021); P95-99 is $233,000 to $630,000; the top 1% is above. Share = group income ÷ total, so P90-99 = top 10% − top 1%.
The history.	High before World War II, low until the 1970s, rising since. Piketty and Saez use tax returns because surveys miss the rich.
Did it rise?	Gap = WH − WL; ratio = WH ÷ WL. Equal dollar raises keep the gap but shrink the ratio.
Other causes, remedies.	Education slowed after 1970 while technology raced on, unions weakened, the real minimum wage fell, and CEO pay soared. Remedies: education (Heckman: early childhood), aid to displaced workers, the EITC (taxpayers pay, jobs rise) or a minimum wage (employers pay, jobs fall).

Trade and trade policy — graphs 5 to 7

Opportunity cost of 1 X = Y given up ÷ X made. From hours per unit, it is hours for X ÷ hours for Y. The lower OC exports X.

- Absolute advantage in both goods still leaves gains if OCs differ; equal OCs mean no gain. Scaling both goods equally changes nothing. Sources: factor abundance and an early start.
- Exporters and buyers of imports win; import-competing firms and their workers lose; the country gains. A tariff may add jobs in the protected industry but probably not in total.
- Tariff DWL = ½ × t × (QS2 − QS1) + ½ × t × (QD1 − QD2). Revenue given: set t × (imports at Pw + t) equal to it and solve for t.
- Pw + t at or above P1 ends imports. A quota above free-trade imports, or a tariff on a good the country exports, does nothing.
- A production subsidy causes only loss d, since buyers still pay Pw. Flatter curves mean imports fall more. Convert first: €320 at €0.80 per $1 is $400.

Micro for the multiple choice — graph 8

Markets match MB to MC. A monopolist raises P and cuts Q. Comparative advantage means the lowest OC.

- The opportunity cost of college is tuition plus forgone earnings.
- Buyer stories shift D, seller stories shift S, and the own price moves along. With two shifts, only one of P and Q is certain.
- A binding ceiling sells Qs (shortage Qd − Qs); a binding floor sells Qd.
- Elasticity = %ΔQ ÷ %ΔP; above 1 in size is elastic. If inelastic, P and spending move together.
- The less elastic side bears more of a tax, whoever pays it in. Taxing inelastic goods gives a smaller DWL.
- Competitive firm: P = MR = MC; profits bring entry until P = ATC. Monopoly MR is the change in total revenue. With no fixed cost, profit = revenue − the area under MC.
- Positive externality: too little, so subsidize. A polluting monopolist: compare Qm with where D meets SMC.
- Rational spending: $MU_a \div P_a = MU_b \div P_b$. Services that need people cannot be automated much, so their MC tracks wages.